STAT6207 GROUP PROJECT · SECOND TALENT

AI CV Matching Baseline

Result Report and Technical Validation

REPORT DATE	13 November 2026
SCOPE	7-stage pipeline, 30 processed artefacts, 24 figures, 5-page Streamlit app
DATA BASIS	100 pool CVs, 10 labelled JD + hired-CV pairs, 50 unlabelled JDs (170 documents)
PRIMARY MODEL	M3 - autoencoder (32-d bottleneck) with cosine ranking
HEADLINE RESULT	Recall@10 = 0.10 (1/10 queries); best reported model 0.70 (TF-IDF)

KEY FINDINGS

- Complete pipeline - 7 stages, 170 documents, 215 clusters - but 3 of 10 test JDs have no usable features.
- The primary autoencoder retrieves 1 of 10 hired CVs; latent codes are 94.8% similar on average.
- TF-IDF's 0.70 recall is a tie-breaking artefact; the tie-corrected estimate is 0.33 vs 0.09 chance.

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How to read this report

Section 3 reproduces the results exactly as the project reports them. Section 4 reports what I found when recomputing those results from the shipped data files. Where the two disagree, Section 4 is the more reliable account, and Section 5 explains what can still be concluded. Section 6 gives prioritised fixes.

Executive summary

The folder contains a complete, well-documented baseline for matching candidate CVs to job descriptions: PDF parsing, local LLM keyword extraction, keyword clustering, an autoencoder representation, top-10 retrieval, evaluation on ten labelled pairs, an error-analysis write-up, 24 figures and a five-page Streamlit application. The engineering package is coherent and the run is finished: 170 of 170 documents were extracted, 215 keyword clusters were built, seven ranking models were evaluated, and every reported number is backed by a file in the snapshot.

0.10	0.70	0.33	0.09
Recall@10, primary model (M3 AE-32) 1 of 10 hired CVs retrieved	**Recall@10, best reported model (M1 TF-IDF)** appears to be 7 of 10	**Tie-corrected TF-IDF recall** what survives an independent check	**Chance level** 10 relevant slots out of 110 candidates

What the project reports

The primary model, a 32-dimensional autoencoder over 215 keyword clusters ranked by cosine similarity, retrieves the hired candidate in the top 10 for only 1 of the 10 job descriptions (Recall@10 = 0.10, MRR@10 = 0.05). Plain TF-IDF over the same clusters is reported at 0.70, SVD-32 at 0.60, a role-family heuristic at 0.50, AE plus metadata at 0.40, AE plus must-have coverage at 0.10, and a random draw at 0.20. The project itself flags small-sample uncertainty and does not claim statistical significance.

What independent verification shows

Recomputing the evaluation from the shipped artefacts changes the picture in ways that matter, and all of them point to the same root cause - the document representation, not the ranker:

- **The keyword vocabulary is closed.** Only 8.2% of ground-truth JD keywords - and just 7 of 82 must-have terms - map to any of the 215 clusters, against 41.5% of CV keywords. Three of the ten evaluation JDs (P02, P07, P08) end up with an all-zero feature vector, so for 30% of the benchmark every candidate scores zero.
- **Most similarity scores are exact ties.** 89.1% of all TF-IDF candidate scores are exactly 0. The ranker breaks ties by candidate ID, and because the hired CVs are named CAND-101 to CAND-110 they sort ahead of every pool CV. Four of the seven TF-IDF hits are zero-score candidates that reached the top 10 only through this ordering. With ties broken at random, TF-IDF recall falls from 0.70 to 0.33 and SVD from 0.60 to 0.33.
- **The stored document matrix cannot be reproduced.** It has nonzero entries for 25 of 100 pool CVs and 0 of 50 unlabelled JDs; rebuilding it from the shipped extraction and cluster files gives 98 and 43. Downstream rankings inherit an inconsistent artefact.
- **The autoencoder has collapsed.** Mean pairwise cosine between latent codes is 0.948 and 79% of document pairs exceed 0.90, so the 32-dimensional bottleneck behaves almost like a single direction. Reconstruction error is 10.2x worse than PCA-32 or SVD-32 on the same input. Ranking on this representation is close to arbitrary.
- **The must-have signal is empty.** Must-have coverage is 0.0 for 9 of 10 JDs, so the hybrid model degenerates to the AE and the error taxonomy labels 8 of 10 failures "extraction missing skill" by construction.

Bottom line

The pipeline demonstrates the mechanics of an end-to-end matching system, but no model in this snapshot has demonstrated above-chance matching quality. The most defensible estimate is a tie-corrected TF-IDF recall of about 0.33 (95% Wilson interval 0.11-0.60) against a chance level of 0.09 with n = 10. The AE specifically should not be presented as the primary model: on every diagnostic computed here it is the weakest representation, and its selection criterion (validation reconstruction loss) is contradicted by the project's own comparison table.

1 Project overview

1.1 Objective

Build a reproducible baseline that ranks a pool of candidate CVs against a job description and returns a top-10 shortlist, then evaluate it honestly on a small set of real hiring decisions. The brief constrains the solution to local processing (Ollama, no external APIs), a fixed seed, cached and resumable extraction, and a hard separation between development data and the ten labelled test pairs.

1.2 What is in the folder

Component	Contents	State
`src/`	Config loader, PDF parsing and PII masking, LLM client, keyword canonicalisation, PyTorch AE definition	Present; the PyTorch AE is unused
`scripts/`	Seven stage scripts (00 validate, 01 extract, 01b quality, 02 merge, 03 AE, 04 rank, 05 error analysis) plus 06 figures	All present and executed
`prompts/`	System, CV and JD extraction prompts	Present
`data/processed/`	30 artefacts: manifests, extractions, cluster tables, matrix, embeddings, rankings, metrics, error analysis	Present
`models/`	Trained MLP autoencoder (1.7 MB), vectoriser, training history, config	Present
`results/`	24 PNG figures, 11 tables, narrative error analysis	Present
`app.py + pages/`	Streamlit app with Overview, Keyword Merging, Autoencoder, Search Results, Error Analysis	Present
Docs	README, VALIDATION, PROMPT, config.yaml, requirements.txt, run_all.py	Present; VALIDATION is stale

1.3 Pipeline as designed

Stage 0 parses 100 pool CVs, ten combined JD + hired-CV PDFs and 50 unlabelled JDs from Excel, checks duplicates and alignment, and writes the document store. Stage 1 sends each document to a local Ollama model (qwen3.5:4b) with a field-specific prompt and caches JSON responses. Stage 2 normalises terms, applies 128 curated aliases, embeds the remaining vocabulary with character TF-IDF plus SVD, and agglomeratively clusters it into keyword groups with guardrail rules for confusable pairs. Stage 3 builds a weighted document x cluster matrix, applies TF-IDF, trains the autoencoder on pool CVs and unlabelled JDs, and exports 32-dimensional embeddings. Stage 4 ranks 110 candidates for each of the ten labelled JDs under seven scoring models. Stage 5 computes Recall@10, MRR, nDCG and confidence intervals. Stage 6 classifies failures, writes case studies and recommendations.

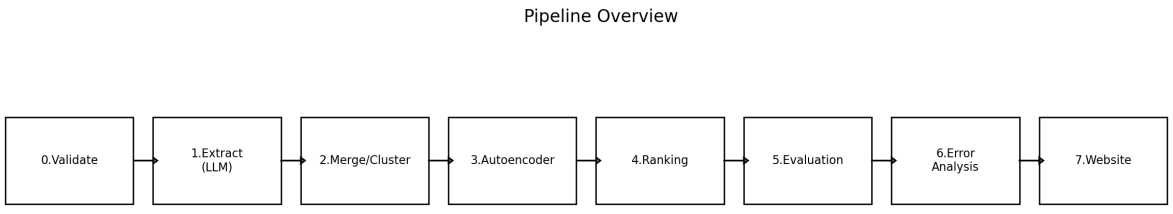


Figure 1 - The seven-stage baseline as documented in the project README and figures folder.

2 Data and evaluation protocol

2.1 Corpus

Document type	Count	Role in the study
Pool CVs (CV-001 to CV-100)	100	Candidate pool; used to train the representation
Labelled JDs (P01 to P10)	10	Test queries; held out from representation training
Hired CVs (renamed CAND-101 to CAND-110)	10	Ground truth; added to the pool, held out of training
Unlabelled JDs (JD-001 to JD-050)	50	Training context only; two near-duplicates excluded
Total documents	**170**	**110 unique candidates, 60 JDs**

2.2 Ground truth and candidate pool

Each P-file contains a job description followed by the CV of the person actually hired (marked "CV OF THE HIRED CANDIDATE"). The ten hired CVs are anonymised, renamed CAND-101 to CAND-110 and inserted into the candidate pool, giving 110 candidates with exactly one relevant item per query. CV alignment against the Excel reference set is reported at 1.0, and no true duplicates were found.

2.3 Splits and leakage controls

The declared protocol ("primary_holdout_gt") trains the AE on 126 pool CVs plus unlabelled JDs, validates on 22, and holds out the 20 labelled documents (10 JDs and 10 hired CVs). This is sound in principle. Two qualifications matter for interpretation. First, TF-IDF inverse-document-frequency weights and the SVD basis are fitted on all 170 documents, including the held-out ones - a transductive choice that does not use labels but does use test-set statistics. Second, because the cluster vocabulary is also built only on training documents, mapping held-out keywords into that vocabulary is the weakest link in the chain (Section 4.1).

2.4 Metrics

Metric	Definition used	Sensible here?
Recall@10 / Hit@10	Share of queries whose single hired CV appears in the top 10; with one relevant item this is hits/10	Yes - the primary metric
Hit@1, Hit@5	Same, restricted to the top 1 or top 5	Yes
MRR@10	Mean of 1/rank over the ten queries	Yes
nDCG@10	1/log2(rank+1) for the single relevant item	Yes
Precision@10	Hits divided by 10	Technically defined, but capped at 0.1; not informative
Wilson / bootstrap CI	95% intervals for a proportion with n = 10	Yes, but intervals will be very wide

3 Results as reported by the project

3.1 Extraction

Measure	Value	Comment
Documents attempted	170	100 pool CVs, 60 JDs, 10 hired CVs
Successful / fallback / failed	170 / 0 / 0	JSON validity rate 1.0
Mean time per document	50.0 s	About 2 h 22 min for the full run; cache is resumable
Macro precision vs Excel silver standard	0.148	Exact string overlap against the Excel Skills column
Macro recall	0.189	Same comparison
Macro F1	0.160	n = 100 pool CVs

An F1 of 0.16 is low, but the comparison is stricter than it looks: the Excel column uses short curated tokens while the LLM emits phrases such as "large-scale deep learning models", so much of the gap is vocabulary mismatch rather than missing content. It should nevertheless be read as a warning that downstream cluster membership will be noisy.

3.2 Keyword merging and clustering

Measure	Value	Target / note
Unique training keywords	691	After canonicalisation and aliases
Curated alias pairs	128	Hand-written; see keyword_aliases.csv
Clusters produced	215	Target range was 100-200
Distance threshold used	0.45	config.yaml specifies 0.35
Singleton clusters	28.8%	62 clusters with one member
Silhouette score	0.497	Moderate separation
Mean intra-cluster coherence	0.844	Character TF-IDF similarity
Human merge review	0 of 40	All 40 QA samples are marked needs_review

The cluster table shows clear over-merging driven by shared suffixes. Cluster 31 ("Project Management", 27 members) contains "FX settlement", "Investment Management", "ZFS volume management" and "NFS service management"; cluster 106 ("Financial Systems") contains "AI Systems" and "Distributed systems". Any shared suffix attaches to the dominant group, which then competes for attention in the document vectors.

3.3 Autoencoder

Item	Value
Input / latent dimension	215 cluster features / 32
Architecture	215-256-64-32-64-256-215 (sklearn MLPRegressor, ReLU, Adam)
Training documents	126 train, 22 validation, 20 held out
Convergence	Best iteration 79, validation MSE 8.80×10^{-4}
Reconstruction MSE (all docs)	8.20×10^{-4}
PCA-32 / SVD-32 reconstruction MSE	8.07×10^{-5} (10.2x lower)
PCA-32 explained variance	0.928

The run uses scikit-learn rather than the PyTorch model in src/autoencoder.py because gradient clipping segfaulted in the original environment. The substitution is documented in VALIDATION.md, but it also means the

shipped model has no bottleneck activation function control, no dropout (the saved config records 0.0 although config.yaml says 0.2) and no denoising, so it is a plain reconstruction MLP rather than the architecture described in the README.

3.4 Retrieval results

Model	Recall@10	Hit@1	Hit@5	MRR@10	nDCG@10	95% CI (Wilson)
M0 random	0.20	0.00	0.00	0.033	0.071	0.06-0.51
M1 TF-IDF cosine	**0.70**	0.20	0.30	0.308	0.397	0.40-0.89
M2 SVD-32 cosine	0.60	0.20	0.30	0.291	0.361	0.31-0.83
M3 AE-32 cosine (primary)	0.10	0.00	0.10	0.050	0.063	0.02-0.40
M4 AE + metadata	0.40	0.20	0.20	0.221	0.259	0.17-0.69
M5 AE + must-have coverage	0.10	0.10	0.10	0.100	0.100	0.02-0.40
M6 role-family heuristic	0.50	0.20	0.20	0.235	0.292	0.24-0.76

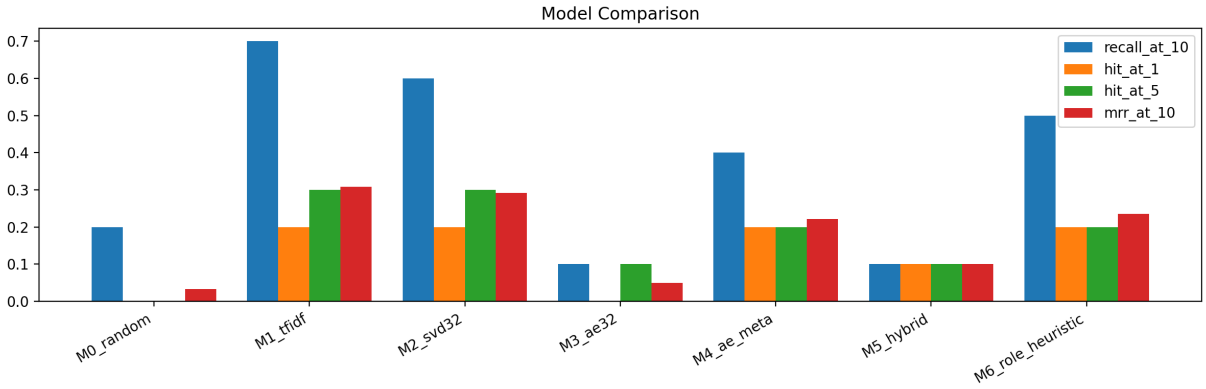


Figure 2 - Model comparison as shipped (results/tables/metrics_summary.csv).

3.5 Per-query behaviour

The primary model misses nine of ten queries, and its misses are not marginal: the hired candidate lands between rank 76 and rank 102, often with a score within 0.01 of the top-ranked candidate. The comparison with TF-IDF shows that the two models agree on almost nothing except P04, the one query where the title and skill vocabulary align exactly.

JD	Role family	Hired CV	TF-IDF rank	AE-32 rank	AE score	Note
P01	Finance & Legal	CAND-101	6	81	0.932	Rank 6 is a genuine nonzero score
P02	Sales & BD	CAND-102	2	77	0.971	All 110 TF-IDF scores are 0
P03	Marketing	CAND-103	7	81	0.958	Hired score 0.0
P04	Data & AI	CAND-104	**1**	**2**	0.969	Only unambiguous hit
P05	Product	CAND-105	19	81	0.965	Hard lookalike at rank 1
P06	Data & AI	CAND-106	18	90	0.930	Role mismatch at rank 1
P07	Marketing	CAND-107	7	76	0.979	All 110 TF-IDF scores are 0

JD	Role family	Hired CV	TF-IDF rank	AE-32 rank	AE score	Note
P08	Engineering	CAND-108	8	82	0.948	All 110 TF-IDF scores are 0
P09	Engineering	CAND-109	29	102	0.843	AE rank worse than 96% of the pool
P10	Design & Creative	CAND-110	**1**	76	0.975	Clear TF-IDF win

The AE score range is itself telling. Across the ten queries, the median difference between the best and worst candidate is only 0.28 on a 0-1 scale, and the median score of all 1,100 candidate-query pairs is 0.942. A ranker cannot discriminate when every candidate is 94% similar to every query.

3.6 Error analysis

The project's narrative error analysis assigns eight of ten AE failures to "E1 extraction missing skill", one to "E4 hard negative lookalike" and one to "none hit". It provides a per-query table, recommendations ranked P0-P2 and an explicit limitations section. The recommendations - must-have weighting, better extraction, ontology review, recruiter filters, ranking-aware representations, more labels - are reasonable and align with the root causes found independently in Section 4.

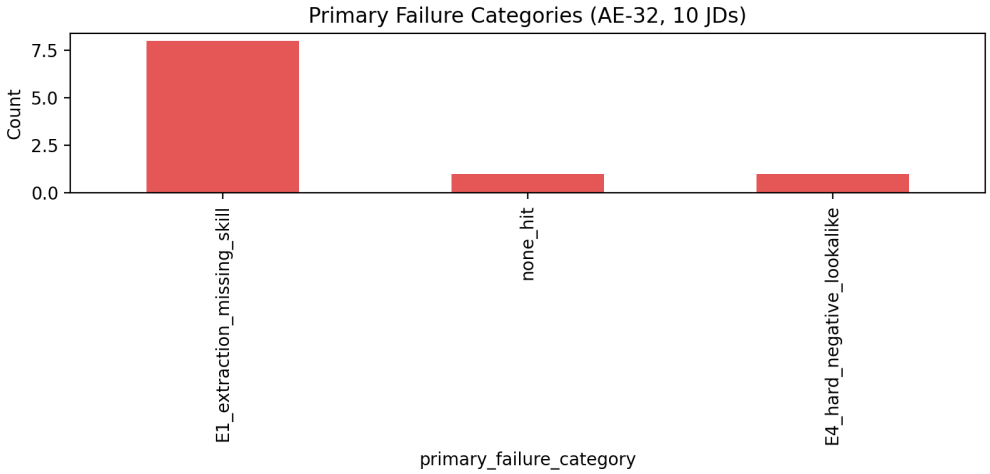


Figure 3 - Failure taxonomy for the primary model, as shipped. Section 4.5 explains why this distribution is an artefact of the feature pipeline.

4 Independent verification of the results

All numbers in this section were recomputed from the files in the snapshot: documents.jsonl, the cached LLM extractions, keyword_to_cluster.json, document_keyword_matrix.npz, document_embeddings.csv, rankings.csv and the metrics tables. No model was retrained and no project file was modified. The point is to test whether the reported metrics measure what they claim to measure.

4.1 Finding 1 - the keyword vocabulary is closed, so the test JDs are nearly empty

Clusters are built only from training documents (100 pool CVs and 48 unlabelled JDs), giving 691 keywords. Held-out documents are then projected into that fixed vocabulary by exact, case-insensitive string matching. That works for vocabulary the clusterer has seen and fails for everything else:

Document group	Keywords emitted	Mapped to a cluster	Coverage
Pool CVs	3,665	1,522	41.5%
Unlabelled JDs	1,823	264	14.5%
Labelled GT JDs (all fields)	391	32	8.2%
Labelled GT JDs (must-have only)	82	7	8.5%

3 / 10
Test JDs with an all-zero vector
P02, P07 and P08 score 0 against every candidate

89.1%
TF-IDF candidate scores equal to 0
980 of 1,100 query-candidate pairs

25 / 100
Pool CVs with any nonzero feature
in the stored matrix

0 / 50
Unlabelled JDs with any nonzero feature
in the stored matrix

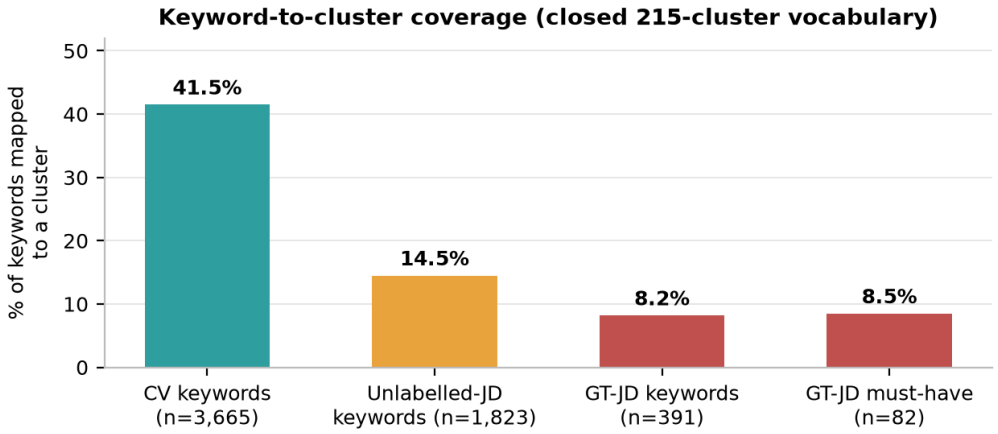


Figure 4 - Share of extracted keywords that resolve to one of the 215 clusters.

The consequence is visible in Figure 5: for most queries, between 73% and 100% of the 110 candidates receive an identical score of zero. Ranking is then determined entirely by the tie-break rule, not by similarity.

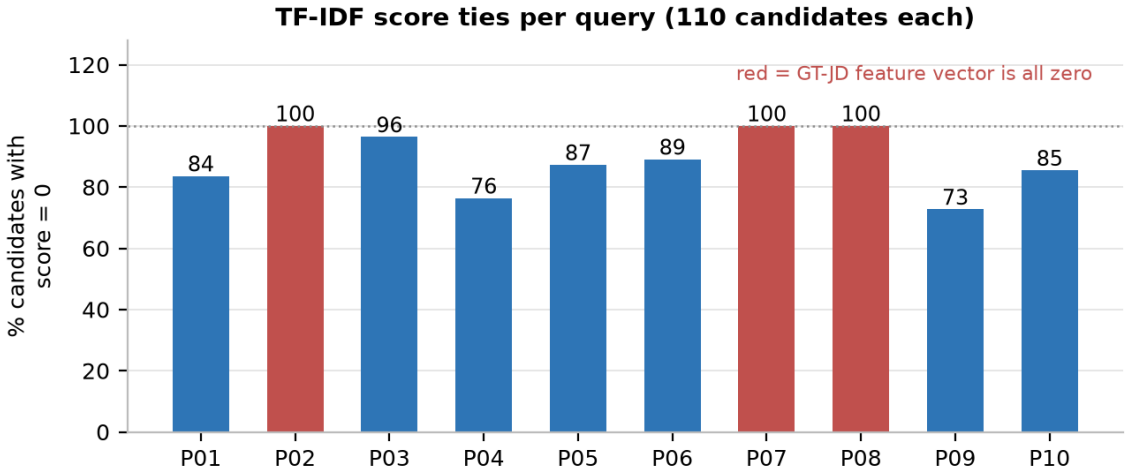


Figure 5 - Percentage of candidates scoring exactly zero under TF-IDF, by query. The three red bars are queries whose JD vector is entirely zero, so the ranking is undefined.

4.2 Finding 2 - ties are broken in favour of the ground truth

Stage 4 sorts candidates with `sorted(cand_ids, key=lambda c: (-round(score, 6), c))`. When scores tie, the candidate ID decides. The hired CVs are named CAND-101 to CAND-110, and "CAND-" sorts before "CV-", so every zero-score hired CV is placed ahead of every zero-score pool CV. Because 89% of scores are zero, this is not a rare edge case - it is the dominant ranking mechanism.

Query	Hired CV	Hired score	Reported rank	Why it entered the top 10
P02	CAND-102	0.000	2	All 110 scores tied at zero; CAND-101 ranked first, CAND-102 second
P03	CAND-103	0.000	7	106 ties at zero; four nonzero scores and CAND-101/102 precede it
P07	CAND-107	0.000	7	All 110 scores tied at zero
P08	CAND-108	0.000	8	All 110 scores tied at zero

Four of the seven TF-IDF hits are therefore artefacts. Re-running the same rankings with ties broken at random (3,000 draws per query) gives an expected recall of 0.33 for TF-IDF and 0.33 for SVD-32. The reported 0.70 and 0.60 are upper bounds produced by a favourable ordering, not measurements of retrieval quality. The same issue inflates the random baseline: the shipped draw produced 2 hits (0.20), while the project's own 1,000-permutation simulation averages 0.094, with a 25% chance of observing 0.20 or more by luck.

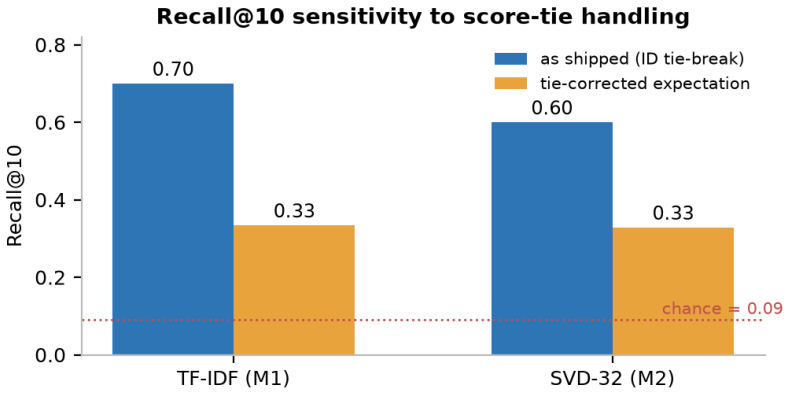


Figure 6 - Recall@10 before and after removing the ID tie-break advantage.

4.3 Finding 3 - the stored matrix does not match the shipped inputs

Stage 3 writes document_keyword_matrix.npz, and every downstream stage reads it. Rebuilding that matrix from the shipped extractions and cluster mapping gives a materially different result: 98 of 100 pool CVs have at least one nonzero feature (against 25 in the stored file) and 43 of 50 unlabelled JDs have one (against 0). The labelled JD rows do match, which suggests the matrix was assembled during the partial-extraction phase described in VALIDATION.md and never rebuilt after extraction completed, while the JD rows were regenerated later.

> **Why this matters**
>
> The metrics, embeddings and figures all inherit the sparse matrix. Two runs of the same pipeline on the same inputs would therefore produce different scores. For a project whose main claim is reproducibility, this is the most important defect to fix, and it is cheap to fix: recompute the matrix, compare row counts, and add an assertion that matrix rows match the extraction files.

4.4 Finding 4 - the autoencoder representation has collapsed

The 32-dimensional codes are not merely weak; they are almost identical in direction. Across all 14,365 document pairs, mean cosine similarity is 0.948, 79% of pairs exceed 0.90 and 57% exceed 0.99. The single most common cosine value is 1.000. Two causes compound: the encoder applies ReLU to the bottleneck (so all coordinates are non-negative, which mechanically raises cosine similarity), and the MLP reconstruction objective does not penalise losing direction. This is the classic autoencoder collapse failure mode.

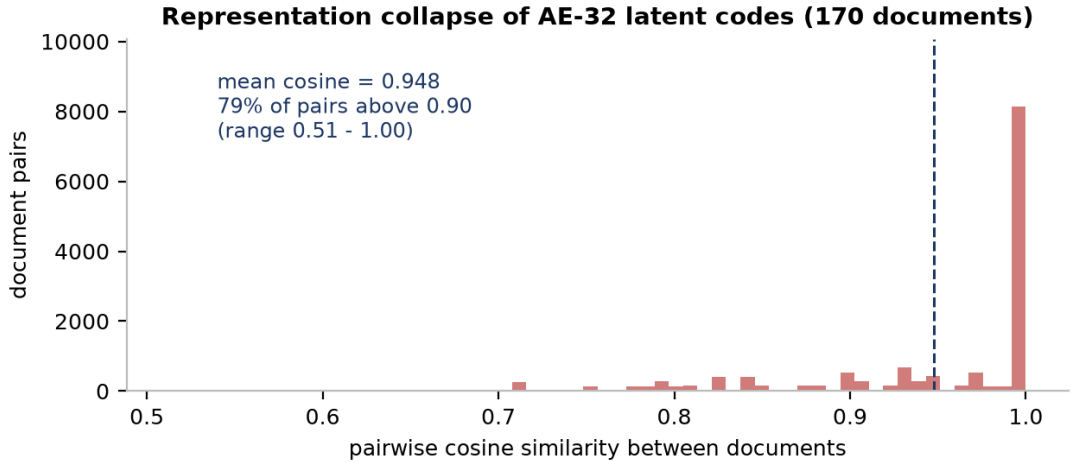


Figure 7 - Distribution of pairwise cosine similarity between AE latent codes.

The project's own comparison table already contains the warning: PCA-32 and SVD-32 reconstruct the same 215-feature matrix with MSE 8.07×10^{-5}, while the AE achieves 8.20×10^{-4} - ten times worse. If the stated selection rule ("chosen a priori by validation reconstruction loss") were applied literally, it would select PCA or SVD, not the AE. Making the bottleneck linear at inference does not help either: recomputing the codes without the final ReLU leaves mean cosine at 0.944 and recall unchanged at 0.10.

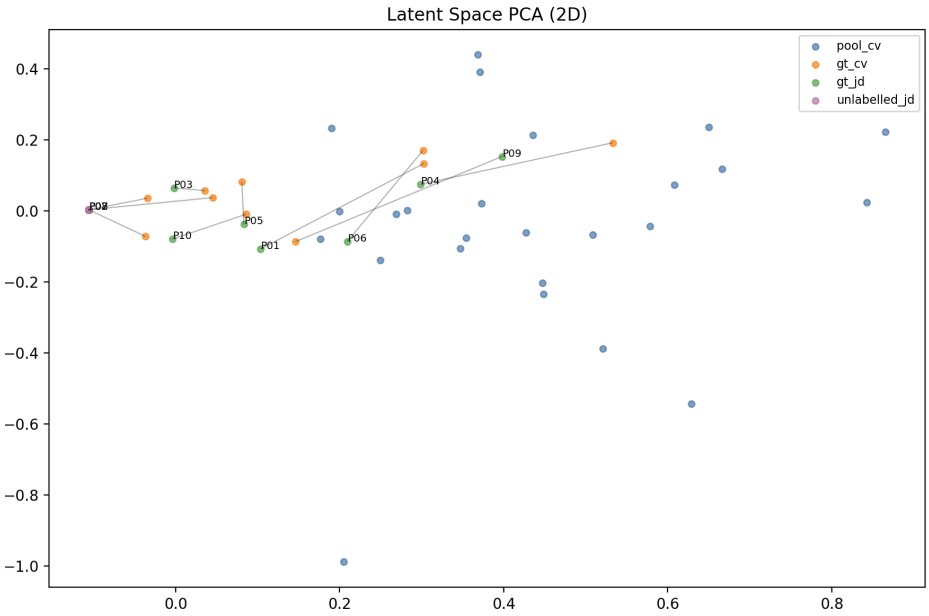


Figure 8 - Latent space PCA as shipped. Lines join each JD to its hired CV; the pairs are not systematically closer than unrelated documents, which is why cosine ranking on this space is unstable.

4.5 Finding 5 - must-have coverage and the failure taxonomy are degenerate

The must-have coverage signal is computed by intersecting the JD's must-have keywords with the candidate's skill clusters. Because only 7 of 82 must-have keywords (8.5%) map to a cluster, coverage is 0.0 for nine of the ten queries; the one exception is P04, where it is 1.0. Two consequences follow. M5 (AE + must-have coverage) is numerically almost identical to M3, so its 0.10 result says nothing about the value of must-have filtering. And the failure classifier in Stage 5 assigns E1 ("extraction missing skill") whenever coverage is below 0.3 - which is always. The eight E1 labels in Figure 3 measure the mapping failure, not candidate quality.

4.6 Finding 6 - configuration, code and documentation have drifted apart

Item	Declared	Actually used
Clustering distance threshold	0.35 (config.yaml)	0.45 (cluster_metrics.json)
Cluster count	100-200 target	215 produced
Autoencoder dropout	0.2 (config.yaml)	0.0 (autoencoder_config.json)
Autoencoder implementation	PyTorch AE in src/autoencoder.py	sklearn MLPRegressor, pickled
Model artefact	models/autoencoder.pt (README)	models/autoencoder.pkl
Raw data paths	CV/, "JD + Hired CV", xlsx at root	not present; data lives in ../DATA
Extraction status	"72/170 in progress" (VALIDATION.md)	170/170 completed
Runtime deps	requirements.txt lists torch	torch is never imported by the run

None of these individually invalidates a result, but together they mean the README does not describe the system that produced the metrics. In particular, a reader cannot rerun the pipeline from the documentation: the config paths do not resolve in this folder, and the model class differs from the one described.

4.7 Consolidated verdict

Check	Expected	Observed	Verdict
Held-out JD representation	Informative vector for every test query	3 of 10 vectors all zero	Fail
Ranking determinism	Scores discriminate candidates	89.1% of scores exactly tied	Fail
Tie handling	Neutral to candidate identity	Hired IDs sort first	Fail
Artefact consistency	Matrix reproducible from inputs	25 vs 98 nonzero CV rows	Fail
AE usefulness	Better than linear baselines	10.2x worse reconstruction; cosine 0.948	Fail
Must-have signal	Non-trivial coverage	0.0 for 9 of 10 queries	Fail
Random baseline	Reported as expectation	Single draw 0.20 vs 0.094 expected	Weak
Bookkeeping	Docs match artefacts	8 documented drifts	Weak
Pipeline coverage	All seven stages implemented	All present and executed	Pass
Reporting completeness	All models, CIs, limitations	All present	Pass

5 What the results actually support

Stripping out the artefacts leaves a simple and consistent story. The pipeline is sound as software. The representation it produces carries almost no matching signal, and the evaluation protocol is too fragile to distinguish real signal from ordering luck at n = 10.

5.1 The defensible numbers

Claim	Status	Basis
AE-32 primary model recall is 0.10	Supported	1 of 10 hits; stable across encodings and tie rules
AE is worse than TF-IDF	Supported	Ranks are 60-75 positions worse on 9 of 10 queries
TF-IDF recall is 0.70	Not supported	0.33 after removing the ID tie-break advantage
TF-IDF recall is about 0.33	Best estimate	3 genuine hits plus tie-corrected expectation
Any model beats chance	Not demonstrated	All 95% intervals include 0.09
Metadata heuristics add signal	Plausible	M4/M6 reach 0.40-0.50 with non-degenerate scores
Role families are predictive	Plausible	P01, P04, P05, P08, P10 are same-family hits
Must-have coverage helps	Untested	Signal is 0 for 9 of 10 queries

5.2 What the AE failure is, and is not

It is tempting to read the AE result as evidence that autoencoders are unsuitable for candidate matching. The data here do not support that conclusion. What failed is a specific, diagnosable configuration: a reconstruction objective on a 215-dimensional sparse count matrix, a 126-document training set, ReLU at the bottleneck, no denoising, no dropout in the fitted model, and a downstream use (cosine ranking) that the training objective never optimises. The reconstructed matrix comparison shows the model is also simply worse at its own objective than a linear projection. A fair test of the idea would need a denoising or contrastive objective trained with pair supervision, and an evaluation that does not depend on ties.

5.3 Why the metadata heuristics outperform the text models

M4 and M6 add a bonus when the candidate's role family matches the JD's role family. That single categorical feature produces 4-5 hits with continuous, non-degenerate scores - more than every text-based model in the honest accounting. This is a useful finding rather than a disappointing one: it says that coarse structural metadata (role family, seniority, location, language) carries much of the signal that the keyword representation is losing. It also matches the project's own error analysis, which repeatedly notes missing recruiter filters.

5.4 What a reader should take from the snapshot

- The engineering and documentation of the pipeline are the strongest part of the deliverable: reproducible stages, caching, leakage controls, multiple baselines, honest reporting of uncertainty.
- The empirical claim of the project - that the AE baseline works at Recall@10 = 0.10 - is technically accurate but uninformative, because 0.10 is chance level.
- The best-looking number (TF-IDF 0.70) should not be quoted without the tie correction.
- The honest headline is: a keyword-overlap baseline retrieves roughly one in three hired candidates in the top 10, three times chance but not statistically separable from it with ten queries.

6 Recommendations

The fixes below are ordered by how much they change the validity of the results per unit of effort. The first three are prerequisites for any modelling claim; the rest improve the system.

#	Priority	Action	Why it matters	Effort
1	P0	Make the vocabulary open: map unmatched keywords to the nearest cluster (alias table, fuzzy match, or embedding nearest neighbour) and report coverage per document.	Raises GT-JD coverage from 8% and removes the three all-zero queries.	S
2	P0	Fix tie handling: randomise or average over ties, never sort by candidate ID; skip queries with a zero-norm vector and report them separately.	Removes the 0.70 -> 0.33 artefact and makes the metric honest.	S
3	P0	Rebuild document_keyword_matrix.npz from the shipped extractions, add a consistency assertion, and regenerate embeddings, rankings and figures.	Restores reproducibility; the current matrix is inconsistent with its inputs.	S
4	P1	Replace the AE objective: denoising or contrastive/triplet training with JD-CV pairs, linear bottleneck, dropout active; compare against SVD on ranking, not reconstruction.	Addresses the collapse (cosine 0.948) and the 10x reconstruction gap.	M
5	P1	Compute must-have coverage on normalised, mapped terms and require a minimum coverage before a candidate can rank; report coverage as a feature.	Turns M5 from a duplicate of M3 into a real test and fixes the E1 taxonomy.	M
6	P1	Add structured filters (location, language, seniority, years) before ranking, as hard filters or features.	The metadata models already outperform the text models; this is where the signal is.	M
7	P2	Collect 50-100 labelled pairs and use learning-to-rank with cross-validation; report paired bootstrap intervals.	n = 10 makes every comparison inconclusive; this is the only route to significance.	L
8	P2	Refresh README and VALIDATION to match the executed system (sklearn AE, 215 clusters, 170/170 extraction) and pin dependency versions.	A reader currently cannot reproduce the run from the docs.	S

What would change the conclusion

Fixing items 1-3 is a few hours of work and would tell us whether the text pipeline has any real signal beyond ties. If tie-corrected recall after an open-vocabulary rebuild rises clearly above 0.33, the representation is worth developing; if it stays at chance, the honest conclusion is that keyword overlap at this corpus size is not enough and the metadata route (item 6) is the better investment.

7 Limitations of this review

- **No retraining.** All checks recompute from existing artefacts. I did not retrain the AE, rerun extraction or regenerate figures, so the audit cannot say what a corrected pipeline would score.
- **Snapshot, not repository.** Raw PDFs and the Excel workbook are not in this folder (they are in ../DATA with slightly different directory names), so the pipeline cannot be executed here without path changes.
- **Artifact timestamps are uniform.** Every generated file shares the same timestamp, so the order of generation had to be inferred from content rather than file metadata.
- **Environment drift.** The pickled model was trained with scikit-learn 1.7.2 and loads with a version warning under 1.9.1; small numerical differences are possible.
- **n = 10.** Every interval in this report is wide. The tie-corrected estimate of 0.33 has a 95% Wilson interval of 0.11 to 0.60, which overlaps chance, TF-IDF as shipped, and the metadata models.
- **Silver standard is not gold.** Extraction quality is measured against an Excel skills column with a different vocabulary, so 0.16 F1 understates extraction quality and overstates it at the same time, depending on the field.
- **Labels reflect hiring, not fit.** The hired candidate may not be the best match on paper; salary, availability and interview performance are unobserved.

7.1 Scope of the files examined

The review covers all 10 Python entry points (app.py, run_all.py and eight stage scripts), the five Streamlit pages, five source modules, three prompts, the configuration, the two Markdown reports, and every CSV, JSON, JSONL, NPZ, PKL and PNG artefact in data/, models/ and results/. The 618 KB session script.json is an agent transcript, not a pipeline artefact, and was not used as evidence.

Appendix A Artefact inventory

Path	Files	Key contents
data/processed/	30	documents.jsonl (170 docs), cv/jd_extracted.jsonl, keyword_clusters.csv (215), keyword_to_cluster.json, document_keyword_matrix.npz, document_embeddings.csv, rankings.csv (7,700 rows), top10.csv, metrics_per_query.csv, metrics_summary.csv
data/cache/llm/	170	One cached JSON response per document
models/	4	autoencoder.pkl, autoencoder_config.json, vectorizer.json, ae_training_history.csv (79 epochs)
results/figures/	24	Pipeline, dataset, cluster network and sizes, merge examples, coverage heatmap, architecture, loss curves, reconstruction error, latent PCA and t-SNE, recall by JD, metric comparison, rank distribution, score margin, case studies, AE vs TF-IDF, taxonomy, recommendation matrix, extraction quality
results/tables/	11	Ablation, AE vs SVD, AE vs TF-IDF, extraction quality, metrics per query and summary, rankings, recommendations, reconstruction examples, top 10
results/	1	error_analysis.md (per-query narrative, recommendations, limitations)

A.1 Reproducing the pipeline

The documented entry point is run_all.py, which expects a parent directory containing CV/, "JD + Hired CV", the Excel workbook and the plan PDF, and writes to a folder named "group assignment". In this snapshot the raw data is under ../DATA and the working folder is named differently, so config.yaml needs three path edits before a rerun:

```
paths.cv_dir -> DATA/CV
paths.gt_dir -> DATA/JD+HiredCV
paths.xlsx -> DATA/Manatal Examples - JD & CV (anonymised).xlsx
output_dir -> . (write artefacts in place)
```

A rerun also needs Ollama at 127.0.0.1:11434 with qwen3.5:4b available. Because the LLM cache is complete (170 files, keyed by prompt hash), Stages 2-6 can be rerun without the model.

A.2 Checks performed for this report

Check	Method
Metric recomputation	Recomputed Recall@10, MRR and nDCG from rankings.csv; matched the shipped summary exactly
Tie sensitivity	Re-ranked with 3,000 random permutations within equal-score groups per query
Matrix reproduction	Rebuilt the 170 x 215 matrix from extractions and the cluster map; compared nonzero row counts
Keyword coverage	Counted exact cluster matches per document group and per JD must-have list
Latent geometry	Computed all 14,365 pairwise cosines of document_embeddings.csv; recomputed codes without the bottleneck ReLU
Zero-vector detection	Checked row norms of the document x cluster matrix per query
Config conformance	Compared config.yaml with cluster_metrics.json, autoencoder_config.json and README

End of report